Reaction Process of Two-step Catalysation Pre-treatment for Electroless Plating on Non-conducting Substrates

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Reaction Process of Two-step Catalysation Pre-treatment for Electroless Plating on Non-conducting Substrates

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SUMMARY – This paper describes the formation process and chemical state of adsorbates produced during the two-step catalysation pre-treatment prior to electroless deposition. Tin species adsorbed during sensitisation are divided into two types, namely one that adsorbs weakly and one that adsorbs strongly. The former exists in equilibrium with Sn^{2+} in the sensitising solution. It easily desorbed from the substrate during the following HCl immersion, and the loss is recovered during an ensuing sensitising step. The latter is not desorbed during the HCl immersion step. A dry process after sensitisation, which should promote oxidation, induces the transition from Sn^{2+} to Sn^{4+}. In the activating step, two reactions proceed independently of each other; that is, an oxidation-reduction reaction between Sn^{2+} and Pd^{2+} takes place, which produces Pd^0 and Sn^{4+}. Simultaneously, some Sn^{2+} is desorbed from the substrate. XPS analysis confirms that tin oxide (or hydroxide) and metallic palladium are the final products in the two-step catalysation pre-treatment.

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INTRODUCTION

Electroless plating has a particular advantage in that it produces a film of uniform thickness across a wide area on non-conductive substrates. However, the application of this method to these substrates requires a substrate preparation process in which catalytic nuclei form on the surface to initiate electroless deposition. The most common substrate preparation process comprises two-steps in which first, the surface is immersed in a tin(II) chloride solution, then in a palladium chloride solution[1–3]. The former is a sensitising step and the latter an activating step. The widely accepted basic reaction is as follows[1].

$$Sn^{2+}_{sol} \rightarrow Sn^{2+}_{ads} \tag{1}$$

$$Sn^{2+}_{ads} + Pd^{2+}_{sol} \rightarrow Pd^{0}_{ads} + Sn^{4+}_{sol} \tag{2}$$

where the terms "sol" and "ads" denote the ions in "solution" and the ions "adsorbed" on to the substrate, respectively. In the second reaction (2), the divalent tin (Sn^{2+}) ions reduce the palladium ions, and nuclei of metallic palladium (Pd^{0}_{ads}) are produced upon the substrate, while the adsorbed Sn^{2+} ions (Sn^{2+}_{ads}) are oxidised to Sn^{4+} and are desorbed from the substrate. Besides the above formulae, several other reaction mechanisms have been proposed for the process[4–14]. For the sensitising step, the state of tin ions in the sensitising solutions[4,5], the ageing of the solutions[6–8] and the amount and morphology of adsorbed tin species[6,9–13] have been discussed. However, opinions on this step are still divided. For the activating step, the above formula (2) is recognised as an advantageous mechanism, while discussions of an hypothesis that Pd–Sn complexes participate in the reaction are fairly often encountered[9,11,13,14]. The authors have investigated the amount and morphology of the adsorbates[16–18], and also the deterioration of the sensitising solution during ageing[19]. In this study, the formation process and the change in chemical state of adsorbates during the two-step catalysation pre-treatment is clarified as part of an ongoing research programme to elucidate the mechanism of the preparation process.

EXPERIMENTAL

Soda lime glass sheets (Asahi Techno Glass Co. Ltd) were used for the substrate material. Prior to the preparation process, the sheets were carefully cleaned, *via* immersion in a 10% KOH solution and then neutralised with a dilute HCl solution, followed by rinsing with distilled water. The preparation process comprises three steps, namely, an HCl immersion step, a sensitisation step and an activation step. The chemical composition of each solution and the operating temperature and time of each respective step are summarised in Table 1, where the steps are abbreviated to "HCl", "ST" and "AV". By combining these steps, eleven types of pre-treatment process, denoted A–K in Table 2, are devised. The solutions were prepared using reagent grade chemicals and distilled water. Because the sensitising solutions deteriorate quickly after preparation, the experiments were carried out within 2 h of solution preparation. The chemical content of palladium and tin in the adsorbates were measured using an inductively coupled plasma spectrometer (Seiko Instrument

Table I. Chemical composition, operating temperature, and duration of each step fortwo-step catalysation pre-treatment

	Sensitisation [ST]	*HCl immersion [HCl]*	*Activation [AV]*
$SnCl_2$ / mol m^{-3}	0.89	–	–
$PdCl_2$ / mol m^{-3}	–	–	0.56
KCl / mol m^{-3}	2.4	12.0	12.0
Solution temp. / K	300	300	300
Immersion time / s	120	10	60

Table II. Eleven types of pre-treatment

Pretreatment	*ST*	*Dry**	*HCl*	*AV*	*ST*	*AV*
A	yes	–	–	–	–	–
B	yes	–	–	–	yes	–
C	yes	–	yes	–	–	–
D	yes	–	yes	–	yes	–
E	yes	yes	yes	–	–	–
F	yes	yes	–	–	yes	–
G	yes	–	–	yes	–	–
H	yes	–	yes	yes	–	–
I	yes	yes	–	yes	–	–
J	yes	–	–	yes	yes	–
K	yes	–	–	yes	yes	yes

*Dry: drying with stream of air at room temperature.

Co. Ltd, SPS4000). Prior to this measurement, the adsorbates were dissolved in a mixed acid solution ($HCl:HNO_3:H_2O$ $=4:1:15$). The chemical states of tin and palladium were determined with the aid of X-ray photoelectron spectroscopic analysis (Shimadzu Co. Ltd, ESCA-1000). The binding energies were corrected using the C-1s peak at 284.6 eV as a standard[20].

RESULTS AND DISCUSSION

It was found that the adsorbates formed during the sensitising step were saturated within 100 s after immersion in the sensitising solutions and therefore the immersion time was fixed at 120 s (Ref. 18).

The tin and palladium content in the adsorbates produced during the pretreatments are shown in Table 1(b), and the results are given in Figures 1(a) and (b). The symbols A to K correspond to the same symbols in Table 1(b). On comparing A to F, the following facts become clear: tin content, which represents the amount of adsorbate produced during the sensitising step, remains at a constant value of 1.3×10^{-5} mol m^{-2} after a repetition of the sensitising step (compare A with B). This implies that the amount of adsorbate is not affected by a simple repetition of the sensitising step. In contrast, the HCl immersion step diminishes the amount of adsorbate (compare A with C), and a dry step inserted between the sensitising and HCl immersion steps inhibits the weight drop caused by the HCl immersion step (compare C with E). Then, either a second sensitising step following the HCl immersion step or the dry step doubles the amount (compare C with D, see F).

On the basis of the above experimental results, the following observation may be made about the process of the sensitising step. Tin species adsorbed during the sensitising process are divided into two types, one that adsorbs weakly and one that adsorbs strongly. The former exists in equilibrium with Sn^{2+} in the sensitising solution and adsorbates of this type are easily removed from the substrate at the HCl immersion step and recovered in the ensuing sensitising step. The strongly adsorbed species is not removed during the HCl immersion step. The dry step, which should promote oxidation, induces the transition from Sn (II) to Sn (IV) species.

During the activating process, palladium species are freshly deposited on the substrates, while adsorbed tin species decrease (compare A with G, Figure 1). When the HCl immersion step or the dry step is inserted between the sensitising step and the activating step, the amount of tin species is almost constant before and after the activating step (compare C with H, and A with I in Figure 1(b)). If the sensitising step was repeated after the sensitising step and the successive activating step, *i.e.* carried out twice, the amount of Sn species increased sharply. Furthermore, an activating step following process J induces an increase in the palladium species and a decrease in the tin species (compare J with K in Figure 1 (b)). This tendency is similar to the above results, namely, the comparison between A and G. If the sensitising step and the activating step are repeated alternately, the catalytic activity for the electroless plating reaction can be determined by which step is performed last. If it is the activating step, catalytic activity is obtained. However, if the sensitising step is carried out last, the adsorbates formed during the activating step should be covered with a divalent tin species originating from the sensitising step.

Subsequently, the effect of the composition of the activating solution on the amount of tin species and palladium species in adsorbates was investigated. The results are shown in Figures 2(a) and (b). Figure 2(a) shows the result obtained when changing the concentration of $PdCl_2$ while maintaining the ratio of $HCl/PdCl_2$ constant at 21.4, while Figure 2(b) shows the result obtained when changing the concentration of HCl while keeping the concentration of $PdCl_2$ constant. The term ΔSn represents the weight loss of Sn during activation. As shown, the amount of palladium increases with an increase in $PdCl_2$ concentration when there is a constant $HCl/PdCl_2$ ratio, while it decreases with an increase in HCl concentration. Contrary to this, the amount of tin after the activation step decreases with an increase in $PdCl_2$ and HCl concentrations and eventually settles to $0.6–0.7\times10^{-5}$ mol m^{-2}. This is almost equal to that of process C in Figure 1(a), suggesting that this amount corresponds to that of the adsorbates adsorbed strongly. It also appears that $Pd/\Delta Sn$, which represents the ratio of the amount of adsorbed palladium and the weight loss of tin during the activating step, is not constant.

The above-mentioned quantitative change to the products does not agree with

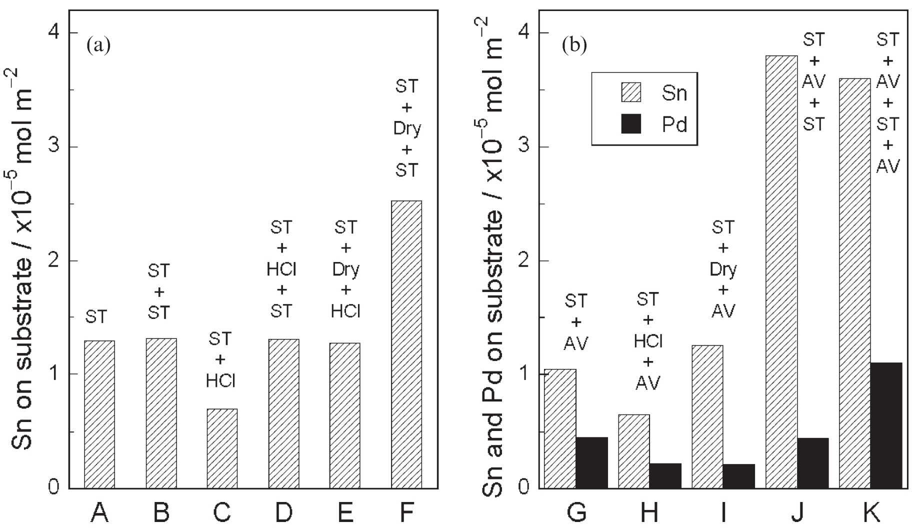

Figure 1. Amounts of Sn and Pd adsorbates on glass substrates. Pre-treatment conditions of A–H correspond to Table 2

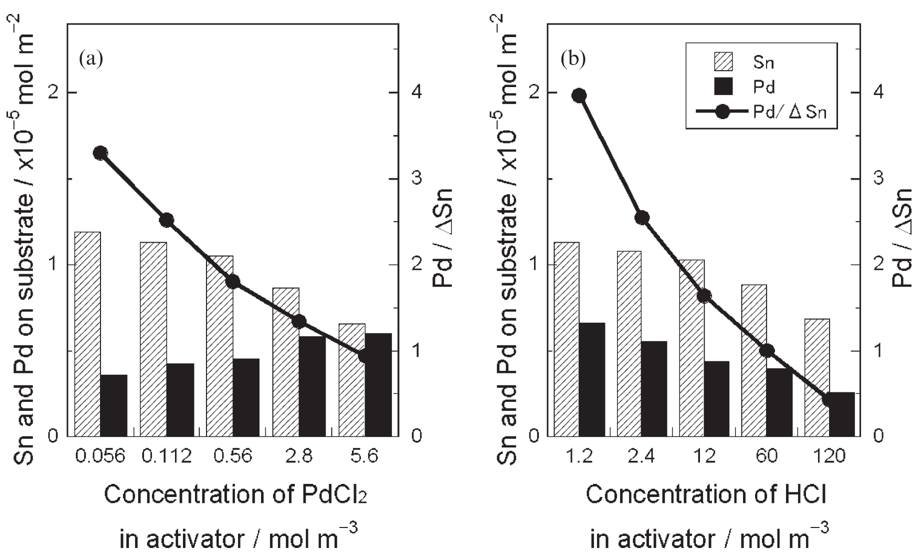

Figure 2. Amounts of Sn and Pd adsorbates on glass substrates and Pd/ΔSn as a function of activator compositions. Regular compositions of activators are (a) HCl/PdCl$_2$ =21.4, (b) PdCl$_2$ =0.56 mol m^{-3}

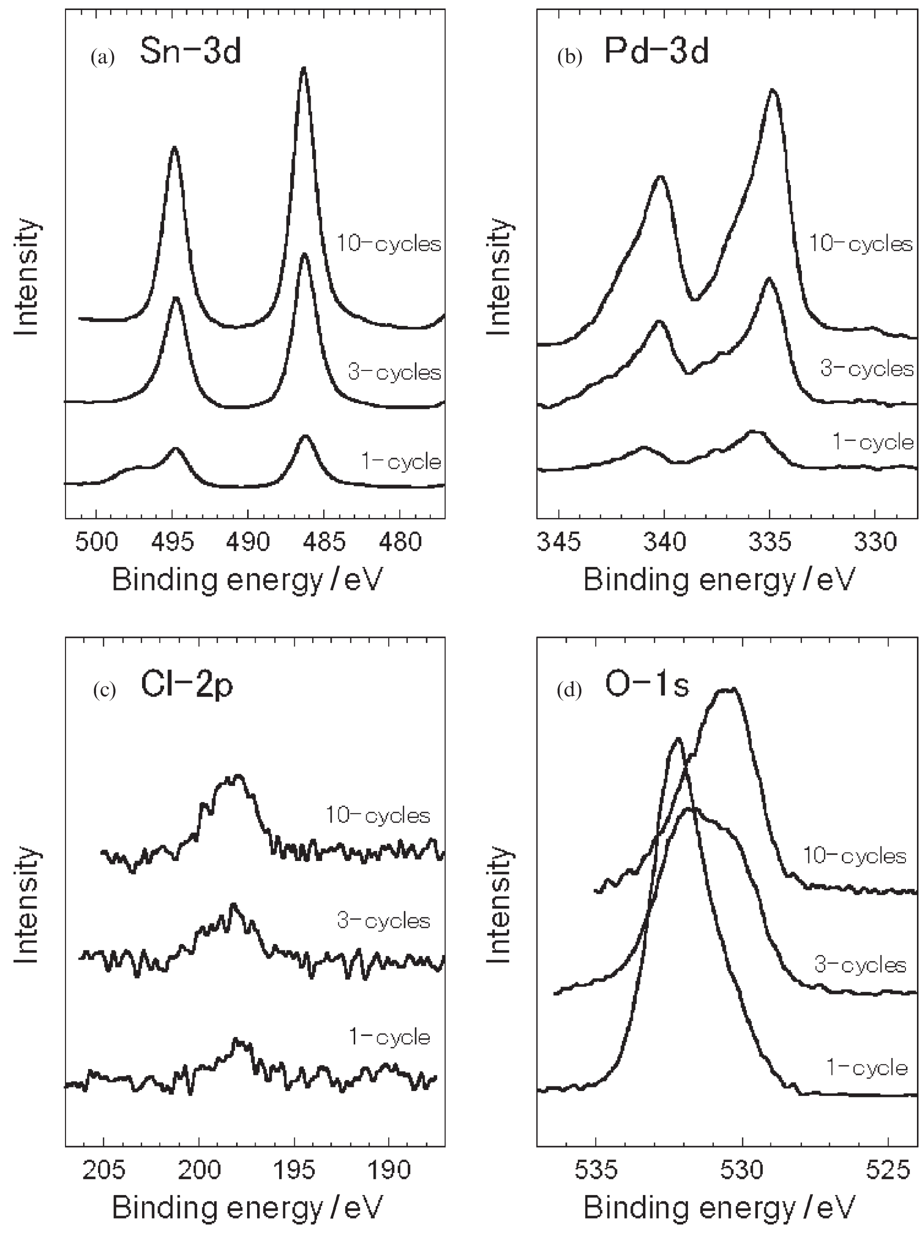

Figure 3. XPS spectra of glass substrates after various pre-treatments and standard materials. (a) Sn-3d, (b) Pd-3d, (c) Cl-2p, (d) O-1s

the values derived theoretically from the reaction shown in equation (2). Furthermore, the composition ratio of palladium to tin in the adsorbates (formed during the activating step) changes, depending upon the conditions of the pre-treatment process. This result reflects the existence of adsorbates with special compositions. These data suggest that some phenomena occur during the pre-treatment process that cannot be explained by the conventional reaction (2). On the basis of the above results, we suggest the following hypothesis. In the activating step, two reactions as shown in equations (3) and (4) proceed independently of each other.

$$Sn^{2+}{}_{ads} + Pd^{2+}{}_{sol} \rightarrow Pd^{0}{}_{ads} + Sn^{4+}{}_{ads} \quad (3)$$

$$Sn^{2+}{}_{ads} \rightarrow Sn^{2+}{}_{sol} \quad (4)$$

The $Sn^{4+}{}_{ads}$ in reaction (3) is insoluble in the activating solution; accordingly $Pd^{0}{}_{ads}$ and $Sn^{4+}{}_{ads}$ are left upon the adsorbate. Simultaneously, a reaction according to (4) takes place. These formulae provide a consistent, full explanation of all the results obtained.

XPS analysis of the adsorbates formed during the pre-treatment was subsequently performed and the results obtained are shown in Figure 3. The pre-treatment was also repeated three and ten times. These repeated treatments will hereinafter be abbreviated as “3-cycle” treatment or “10-cycle” treatment. With the repetition of the pre-treatment, the peak of Sn-3d becomes higher, and its position located at Sn-3d$_{5/2}$ at 484.2 eV does not shift (see Figure 3(a)). The binding energy is higher than that of metallic tin at 484.65 eV (Ref. 20), and close to that of tin oxide or tin hydroxide. The other peak in Figure 3(a) at 494.7 eV corresponds to that of Sn-3d$_{3/2}$.

After one cycle of treatment the binding energy of Pd-3d$_{5/2}$ is 335.6 eV, a value that is higher than that of metallic palladium 334.9 eV (Ref. 19) (see Figure 3(b)). After the 3- or 10-cycle treatment, the peak corresponding to metallic palladium can be observed. In addition, after deconvolution a peak in accordance with that of palladium oxide can be detected. From these data, the existence of a compound that consists of both metallic palladium and palladium oxide is recognised. The quasi-quantitative amount of chlorine is evaluated from the ratio of peak height, namely Cl/Pd and Cl/Sn. The ratios are fairly low, 0.12 and 0.04, respectively. The binding energy of O-1 s after one cycle of treatment is 532.3 eV, which should agree with that of oxygen in the glass substrate. With the increase in cycles, the peak height diminishes, while the other peak at 530.3 eV increases. The latter peak corresponds to tin oxide or tin hydroxide. The above results demonstrated that the adsorbates that were produced during the repeated pre-treatment comprise tin oxide and/or tin hydroxide and metallic palladium.

CONCLUSION

From the above work, the following conclusions may be made.

1. Tin species adsorbed during sensitisation comprise two types, namely a type that adsorbs weakly and one that adsorbs strongly. The former exists in equilibrium with divalent tin (Sn^{2+}) in the sensitising solution. It easily desorbed from the substrate following HCl immersion. The strongly adsorbed species is, however, not desorbed during the HCl immersion step.

2. Two independent reactions proceed during the activation. An oxidation-reduction reaction between Sn^{2+} and Pd^{2+} takes place, and Pd^0 and Sn^{4+} are produced. Simultaneously, a part of the Sn^{2+} detaches from the substrate.

3. Tin oxide (or hydroxide) and metallic palladium are confirmed as the final products in the pre-treatment, using XPS analysis.

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